

Chapter 3: Wave Optics & Quantum Dual Nature

3.1 Wave Optics & Young's Double-Slit Experiment

Interference, Diffraction, and Wavefronts

Christiaan Huygens proposed that every point on an advancing wavefront acts as a secondary source of spherical wavelets. In 1801, Thomas Young proved the wave character of light by shining monochromatic light through two closely spaced parallel slits, producing an alternating pattern of bright constructive fringes and dark destructive fringes on an observation screen.

Constructive interference occurs when path difference $\Delta x = d \cdot \sin(\theta) = m \cdot \lambda$, where m is an integer order.

Key Formula / Law: Double-Slit Fringe Spacing: $y_m = (m \cdot \lambda \cdot L) / d$ | Fringe Width: $\beta = (\lambda \cdot L) / d$

3.2 The Photoelectric Effect & Einstein's Photon Hypothesis

Breakdown of Classical Wave Theory

Classical wave theory predicted that electron emission from a metal irradiated with light should depend upon intensity (amplitude) and exhibit a time lag for low intensities. Experimental observations by Heinrich Hertz and Philipp Lenard completely contradicted this: 1. Emission is instantaneous. 2. Kinetic energy of emitted photoelectrons depends solely on frequency, not intensity. 3. Below a threshold frequency ν_0 , zero photoelectrons are emitted regardless of intensity.

Albert Einstein resolved this in 1905 by postulating light travels in quantized packets of energy called photons: $E = h \cdot \nu$.

Key Formula / Law: Einstein's Photoelectric Equation: $K_{max} = h \cdot \nu - \Phi$ where $\Phi = h \cdot \nu_0$ is the work function of the metal

3.3 Wave-Particle Duality & de Broglie Wavelength

Matter Waves in Quantum Mechanics

Louis de Broglie hypothesized in 1924 that if radiation exhibits both wave and particle characteristics, matter should likewise possess dual wave-particle properties. For any particle possessing momentum $p = mv$, its associated matter wavelength is $\lambda = h / p$.

Key Takeaways:

- Light exhibits wave phenomena during propagation (interference, diffraction) and particle behavior upon interaction/absorption (photoelectric effect).
- Photon momentum is $p = E/c = h/\lambda$.
- Electron diffraction confirmed matter waves experimentally.

